



Mid-Term Examination: 2025

Subject: Web Technology
Level: BCA 3rd Sem

Time: 3:00 Hr

Full Marks: 60
Pass Marks: 30

Group 'B'

(3x5=15)

Attempt Any Three Questions

2. Compare and Contrast between a block-level element and an inline element in HTML? 5
3. Define 3 tier architecture with appropriate diagrams. 5
4. Use Html and CSS to create menubar as given below (use colors on your own) 5



5. Create a form using html and css .

(5)

Student Registration Form

Name:

Email:

Password:

Age:

Gender: ☐ Male ☐ Female ☐ Other

Field of Interest:

Message:

☐ By checking I accept [terms and conditions](#)

6. Write the HTML Tag to generate the following Table

	Average		Red
	Height	Weight	Eyes
Males	1.5	0.004	45%
Females	1.4	0.006	47%

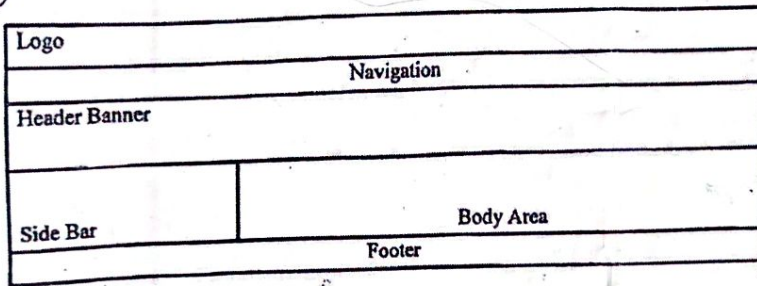
7. What is a CSS selector? Explain different types of CSS selector.
8. Define session. Write its uses. Give an example for using a session.

Group 'C'

Attempt the following question.

(1x10=10)

10. Write a server side script to store form data in the student table which resides under the database called 'saintLawrance'. The structure of the student table (id, name, address, course, phone). Apply validation in form so that it doesn't accept null value.
12. Write a Server side scripting to display all the records stored in the student table which resides under a database called 'student'. The Structure of the student table as (id, name, email, phone, address)
11. Design the following layout of webpage with use of div and appropriate CSS.



***** All the best *****

```

In new CSS
{
display: flex;
flex-direction: row;
gap: 20px;
justify-content: space-between;
}
{
color: white;
}
{
color: green;
}

```

```

In log our CSS
{
side-bar {
display: flex;
height: 200px;
width: 200px;
border: 1px solid black;
flex: 1;
}
body = area {
flex: 2;
}
{
navigation {
text-align: center;
padding: 10px;
}
}

```




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Mid Term Examination 2081

Bachelor in Computer Applications

Course Title: System Analysis & Design

Code No: CACS 203

Semester: III

Full Marks: 60

Pass Marks: 24

Time: 3 hours

Group B

Short Answer Questions (Any SIX)

[6×5=30]

2. What do you understand by system analysis and design? Explain the major roles of a system analyst. [2+3]
3. Explain RAD model. State the advantages and disadvantages of RAD approach. [5]
4. What do you understand by documenting a system? Differentiate between system documentation and user documentation. [2+3]
5. List and describe the steps in the project initiation and planning process. [5]
6. What is system? What are the phases of SDLC? Explain brief? [1+4]
7. What are the ways of managing information project? Explain in detail. [5]
8. Write Short notes on (any 2): [2.5+2.5]
 - a. BPP.
 - b. Gantt Chart.
 - c. Cloud Computing.

Group C

Long Answer Questions (Any TWO)

[2×10=20]

9. What are the major differences between Agile methodologies and waterfall model? Why should you use agile methodologies? Explain. [5+5]
10. Why is the project management important? Describe the concept of integrated CASE tools with its applications. [4+6]
11. List down and discuss the different types of project feasibility factors. Describe Spiral model. [6+4]

Q

customer satisfaction.
Following latest trend
Maintaining format.
Risk management.
information Analyze.

St. Lawrence College

Chabahil, Kathmandu

Mid-Term Examination 2081

Subject: Data Structure and Algorithms

Level: BCA 3rd Sem

Time: 3:00 Hr

Full Marks: 60

Pass Marks: 24

Group-B

Attempt any six questions.

(6 x 5 = 30)

1. ✓ What do you understand by data structure and algorithm? Explain Abstract Datatype.
2. ✓ Explain asymptotic notations of an algorithm.
3. Evaluate the infix expression $A + (((B-C) * (D-E) + F)/G) * (H-I)$ using stack for values $A = 8, B = 4, C = 6, D = 5, E = 10, F = 4, G = 5, H = 7, I = 3$.
4. ✓ What is stack? Write an algorithm or procedure and C function to perform PUSH operation in stack.
5. ✓ Explain different types of queue.
6. Define Recursion. Write a program to find the solution of TOH problem.
7. ✓ What are the different types of linked list? List out the applications of Linked lists.

Group-C

Attempt any two questions.

(2 x 10 = 20)

8. What is Prefix expression? Write an algorithm to convert infix into prefix expression. Trace the following expression into prefix expression. $(A+B*C)+(D-E/F)$
9. ✓ Define Queue. List the applications of queue. Write an algorithm to insert and delete an item in Queue using Linked list.
10. What is circular singly linked list? How does it differ from circular doubly linked list? Write an algorithm or function to add a node at the beginning and end of circular singly linked list.



St. Lawrence College

First Term Examination - 2081

Level: BCA 3rd Semester

Subject: OOP in Java

F.M: 60

P.M: 24

Time: 3 Hours

Group 'B'

Attempt any Six questions:

2. Define java. Explain the different java buzzwords which makes it stand out among other programming languages.
3. Define inheritance. Explain the different types of inheritance with examples.
4. What is encapsulation? What is the need of encapsulation? Illustrate encapsulation with a sample code.
5. What is exception? How does it differ from error?
6. Write a program illustrating exception handling in Java.
7. What is Swing? What are the advantages of Swing over AWT?
8. Define JFrame. Explain the different ways of creating JFrame in swing with example.

Group 'C'

Attempt any Two questions:

9. What is JVM? Explain the working of JVM? How does JVM helps in making java platform independent?
10. Define class and object. Create a Java class Book that represents a book with the following properties: title (String), author (String), price (double) and publishedYear (int). The class should have the following: A constructor to initialize all properties, a method displayDetails() that prints the details of the book and a method isExpensive() that returns true if the price of the book is greater than 500, otherwise returns false. Create an instance of the Book class in the main method, and use the methods to display the details of the book and check if the book is expensive.
11. Differentiate between overriding and overloading? Write a program to illustrate method overriding.

St. Lawrence College
Mid-Term Examination – 2081

Bachelor of Computer Applicatiuon
Course Title: Probability and Statistics
Semester: III

Full Marks: 60
Pass Marks: 24

Time: 3 hours

Group- "B"

Attempt any SIX

[6 X 5=30]

1. Describe scope and limitation of Statistics.
2. The combined mean and variance of salary of 250 workers of city A and city B 560 and 5497 respectively. The mean and variance of the salary of 150 workers of city B are 500 and 81 respectively. Find the variance of salary in city A.
3. Calculate quartile deviation of marks obtained by 20 students in an internal assessment.

Marks	8	9	10	12	15	17	19
No. of students	2	5	5	6	1	1	0

4. Mean of 100 items was 50. Later on it was found that two items were misread as 92 and 8 instead of 192 and 88. Find the correct mean.
5. Calculate coefficient of correlation between two series given below.

X	10	20	30	40	50
Y	3	2	1	5	4

6. Depending upon the data given, which two groups will be easier to teach? Explain with reasons.

Morning BCA Program		Day BCA Program	
Age	No. of students	Age	No. of students
23	9	27	10
29	2	31	8
28	5	30	5
22	10	29	4
30	1	28	6
21	4	33	5
25	11	34	5
26	6	35	11

27	3	36	2
24	9	32	4

Group – “C”

Attempt any TWO questions:

[2*10=20]

1. Following table shows the distribution of participation of Nepalese women in economic activities in 2011 and 2012 (as percentage of total economically active population).

Participated in	2011	2012
Economic activity	46.2	45.8
Labour force	34.6	40.4
Agricultural work	36.4	45.0
Non-agricultural work	16.6	20.3
Manufacturing	14.9	22.9

In which year the distribution of participation of women is variated highly.

2. The advertisement expenses and the sales of a new product are recorded as below.

Adv. Exp	1	5	6	8	10
Sales	50	60	80	100	110

Estimate the sales when the advertising expenses is Rs. 1500.

3. From the following table showing the marks distribution in a BCA class, determine:

- Limits of the marks for the middle 50 % of students.
- The lowest marks obtained by top 40% students.
- The minimum pass marks if 40% students failed the exam.

Marks	0-20	20-40	40-60	60-80	80-100
No. of students	12	18	36	24	10

THANK YOU